

| Question    |     |     | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks Available |     |     |       |       |      |
|-------------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|             |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5<br>F<br>T | (a) | (i) | <p><b>Indicative content:</b> Draw a cross on a square of white paper. Measure (25 cm<sup>3</sup>) of hot sodium thiosulfate using the (25 cm<sup>3</sup>) measuring cylinder and pour into the conical flask. Record the temperature of the solution using a thermometer. Using the 10 cm<sup>3</sup> measuring cylinder, measure out 5cm<sup>3</sup> of the hydrochloric acid. Place the conical flask onto the cross and add the hydrochloric acid. Swirl the flask to mix the contents and at the same time start the stopwatch. Look down at the cross from above the mixture. Stop the stopwatch as soon as the cross disappears. Record the time taken for the cross to disappear.</p> <p>Repeat for the same temperature and then for different temperatures of sodium thiosulfate.</p> <p><b>5– 6 marks</b><br/>Full sequential account of method including measuring instruments and suggested volumes</p> <p><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3 – 4 marks</b><br/>Most steps and measuring instruments included.</p> <p><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Limited method and some instrumentation included. Candidates may have investigated an independent variable other than temperature.</p> <p><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/>No attempt made or no response worthy of credit.</p> | 2               | 4   |     | 6     |       | 6    |

| Question |     |       |  | Marking details                                                                                                                                                                                                                         | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (ii)  |  | Any 2 × (1) from:<br>Same concentration of acid (1)<br>Same concentration of sodium thiosulfate (1)<br>Same volume of acid (1)<br>Same volume of sodium thiosulfate (1)<br><br>Same flask / same cross = neutral                        |                 | 2   |     | 2     |       | 2    |
|          |     | (iii) |  | (A change in concentration/volume) changes the rate of reaction/time to obscure cross                                                                                                                                                   | 1               |     |     | 1     |       |      |
|          | (b) | (i)   |  | As {more water added / less sodium thiosulfate} (concentration of sodium thiosulfate decreases) so {cross takes longer to disappear/decreases rate of reaction} (1)                                                                     |                 |     | 1   | 1     |       | 2    |
|          |     | (ii)  |  | Thiosulfate particles become less crowded/ less thiosulfate particles to react (1)<br>Smaller chance of collisions between (reacting) particles/ lower rate of collision between (reacting) particles (1)<br><br>Accept converse answer | 2               |     |     | 2     |       | 2    |
|          |     |       |  | Question 5 total 12                                                                                                                                                                                                                     | 5               | 6   | 1   | 12    | 0     | 12   |